

SAHAKAR SETU — Grievance Module (Detailed)

How a spoken complaint becomes a formal, addressed, trackable application — in Hindi, without the member writing a single word.

1. The Problem This Solves

A rural PACS member with a stuck PMFBY claim or a loan dispute cannot draft a formal application: they don't know the format, the correct addressee, or the legal language. Sahakar Setu interviews them conversationally (on any channel), extracts the facts, and generates a formal grievance application addressed to the **correct authority** — with a tracking ID they can check from a feature phone.

2. Eight Grievance Categories → Correct Addressee

Category	Addressee (Hindi template)
society_pacs	PACS Secretary / PACS Prabandhak
cooperative_bank	Bank Nodal Grievance Officer
insurance_pmfby	Insurance Grievance Cell / Bima Lokpal
registrar_level	Registrar of Cooperative Societies
election	Registrar / Cooperative Election Authority
loan	Branch Manager, Sahakari Bank
subsidy	District Cooperative Officer
land	Tehsildar / Sub-Divisional Magistrate

3. Fact Memory Feeds the Template

Conversation turns are mined for structured facts (`extractFacts` in `src/modules/case-memory/`): **name, father's name, village, district, state, PACS name, membership number, issue description, application date, receipt number, amount, phone**. Facts accumulate in the case (`case_facts` table) across calls and channels, so the grievance draft fills itself without re-asking. (The code uses camelCase `factKey/factValue` ; Supabase stores snake_case `fact_key/fact_value` — mapped in `getCaseFacts` .)

4. Draft Generation (Handlebars, Hindi-first)

`generateDraft` (`src/modules/grievance/index.ts`) compiles `templates/grievance.hi.hbs` with the facts. A generated Hindi draft looks like:

सेवा में,
श्रीमान सचिव, प्राथमिक कृषि साख समिति (PACS Secretary)

विषय: PMFBY शिकायत: गेहूं की फसल का नुकसान

महोदय,

मैं, राम लाल, ग्राम सरायकेला का सदस्य (सदस्यता सं. 4521) हूँ। मेरी गेहूं की फसल ओलावृष्टि में नष्ट हो गई है। मैंने दिनांक 12-03-2026 को आवेदन किया था, रसीद सं. 88412। अब तक दावा लंबित है।

कृपया उपरोक्त मामले का निस्तारण 15 दिनों के भीतर करने की कृपा करें।

सादर,
राम लाल

दिनांक: 28-09-2026

ट्रैकिंग आईडी: SS-2026-000006

5. Tracking IDs That Survive

- Format: `SS-2026-XXXXXX` (e.g. the live-verified `SS-2026-000006`).
- Collision-safe: ID generation retries up to $3\times$ if the ID already exists in `grievance_drafts` .
- Stored in `grievance_drafts` (`case_id`, `tracking_id`, `addressee`, `subject`, `body`, `language`) + a `grievance_tracking` status row (`drafted`).
- Track on any channel: `GET /api/v1/grievance/track/:trackingId` returns the draft + full status history (newest first).

6. Time-Based Escalation Ladder

`getEscalationPath` (`src/modules/escalation/`) computes the member's current step from the case age, with a ready draft template at every level:

Step	Authority	Triggers	Draft template
1	Society / PACS Secretary	Immediately (status <code>open</code>)	<code>society_grievance</code>
2	Registrar of Cooperative Societies	> 15 days without resolution	<code>registrar_grievance</code>
3	Insurance Ombudsman	PMFBY cases > 30 days	<code>ombudsman_complaint</code>
4	Legal Aid (NALSA)	Escalated cases > 60 days	<code>legal_aid_application</code>

7. Honest Case Strength (Verdict Engine)

- Each case gets a **0–100 strength score** with the missing evidence listed — no false hope.
- Cases scoring **60+** are offered verified-advocate connect (`/api/v1/lawyer/connect`).

- The score comes from facts present vs. facts required for that grievance category (documents, dates, receipts).

8. Kiosk Printout

On the kiosk (Capacitor Android app), the draft and tracking receipt print on a 58/80 mm ESC/POS thermal printer via USB OTG — the member walks away with a signed-form-ready paper application. On the web/PWA the same draft is copyable and downloadable.

Live-verified: grievance draft + tracking flow tested end-to-end on the deployed server — Hindi template fills facts correctly, tracking ID retrievable.